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Letter to the Editor

Stellar feedback in nearby dwarf starburst Mrk 71: does ISM preprocessed with hierarchical star formation resolve LyC leakage timing issue?

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ABSTRACT

Quantifying LyC leakage in the Early Universe is crucial for constraining the timing of the reionization epoch. If the ISM is preprocessed with feedback from earlier star formation, it can open more paths into the CGM for LyC photons from younger populations. Mrk 71, known as the nearest GP analog, demonstrates such staged star formation: the younger embedded cluster Knot A is a primary source of ionizing radiation and LyC the older Knot B has generated a kiloparsec-scale blowout.

We measured kinematics of the outflow originating from Knot B. The warm ionized gas shows moderate supersonic velocities (25-30 km/s, 1.5–3 M). Velocity gradients along and across the outflow are weak. With large spatial extent (1.5–2 kpc), the instantaneous dynamic timescale exceeds 50 Myr, ~ 1.2 ± 0.4 dex more than the age of the cluster. Since no substantially older, RSG-bearing population is observed in Mrk 71, the outflow should have been built by a substantially faster earlier feedback phase.

We show that knot B, modeled as a single stellar population with energy-driven wind, cannot provide enough mechanical energy within its lifetime to build the outflow. The budget can fit if an older knot B periphery (30-50% of the core mass) exists and have provided multiple supernovae. The reported age inconsistency of the cluster, presence of WR population around it and the morphological maturity of the blowout support this interpretation.

We conclude that intra-cluster hierarchical structure can be required to build LyC escape channels within the available stellar-feedback time. Hierarchical star formation may therefore have effect on regulating LyC leakage not only between distinct clusters, but also within individual clusters. However, prevalence of intra-cluster age structure and its role in LyC escape during reionization remain unconstrained.

1. Introduction

The ionizing photon budget of the reionization epoch is among the most important observational constraints provided by the early Universe (Jaskot 2025). Recenty measured UV luminocity functions and ionizing efficiences from JWST observations have revealed tension of $\sim 4\sigma$ between the predicted and observed optical depth of the CMB, which is a significant challenge for the ΛCDM timeline (Muñoz et al. 2024; Melia 2024). The proposed explanations include lower ionizing photon efficiency and/or escape fraction (Begley et al. 2025; Pahl et al. 2026), higher ISM clumping factor (Davies et al. 2024) and increased star formation burstiness (Papovich et al. 2026; Clarke et al. 2024). AGN contribution into the reionization is actively debated too (Madau et al. 2024). None of the proposed solutions does resolve the tension, however, their combination has significantly weakened it.

Quantified tyming of LyC escape is crucial to reproduce time-averaged escape fractions. It was recently shown that the strongest LyC leakers are radiation-feedback dominant, while SN-dominated $f_{\rm esc}$ are lower (Carr et al. 2025). Particularly, while LyC production from an SSP drops after first 3 Myr (reaching ~1 dex at 5 Myr in Starburst99), SN feedback onset in the metal-poor environment should be delayed up to 10 Myr due to weaker winds and greater fraction of failed SNe (Jecmen & Oey 2023). This draws attention to the role of pre-SN feedback in preprocessing the ISM and clearing escape channels. Energy-driven feedback, operating at M~10, can generate narrow chimneys to transport LyC outside the host clouds, but geometry-bounded escape fractions are to be tiny. Simulations provide evidence of significant escape fractions from the host molecular clouds in the first 1-3 Myr (Kim et al. 2019), but LyC escape into the CGM is still poorly constrained. Spatially resolved studies of the z = 4..6 population responsible for reionization are extremely limited, drawing attention to their analogs in the local universe, such as GP galaxies (Cardamone et al. 2009). Markarian 71 (Mrk 71), a giant star-forming region in the nearby dwarf metal-poor galaxy NGC 2366, is well known as a closest analog of GPs (Micheva et al. 2017).

Mrk 71 hosts two young massive clusters (YMCs): a brighter and younger knot A ($M \sim 1.4 \times 10^5 M_\odot$) is ~1 Myr old and is still embedded in gas and dust; it shows the greatest Hα and [O iii] surface brightnesses in the local universe. Knot B is ~10 times smaller and somewhat older (3-5 Myr). It has young bright core (best fitted as a 2.5 Myr SSP) and scattered older periphery (4-5 Myr), which contains an LBV and 3-6 WR candidates Drissen et al. (2000). Knot B hosts a small cavern (R~ 9-14 pc, James et al. (2016)) and has generated an outflow blowout starting from the edge of the cavern, first identified by Roy et al. (1991). van Eymeren et al. (2007, 2009) connected the outflow with filaments seen in Hα image, and compared Hα

and H i velocity fields and have shown (see fig. 7 in the latter paper) that clear redshifted filaments with speeds up to 50 $\mathrm{km\,s^{-1}}$ can be traced in Hα − H i residual velocity map for at least ~ 1.7 kpc in the plane of sight.

The morphologically developed blowout from Knot B lacks any signs of past SN activity (Oey et al. 2023; Kaaret & Prestwich 2024), particularly, its radio spectrum is flat (Chomiuk & Wilcots 2009). This suggests primarily pre-SN feedback, however, several first SN are not ruled out. It provides a low-dencity channel open into CGM at the time the most massive stars still exist, observed in the moment when its LyC leakage should be peaking (Jaskot & Oey 2013; Micheva et al. 2019).

Amorín et al. (2024) demonstrated that the strongest LyC leakers have broader emission lines and more complex line morphologies, while Jaskot et al. (2017) has shown that superwinds are suppressed in the most extreme leakers. The nontrivial connection between LyC leakage and wind velocities makes case study of the outflow in Mrk71 particularly promising.

X-ray source worth mentioning? (Thuan et al. 2014; Kaaret & Prestwich 2024)

To reveal spatial and kinematic structure of the outflow, we performed observations of NGC 2366 in emission lines Hα and [O iii] with scanning FPI of the 6-meter telescope of the Special Astonomical Observatory in Russia. We discuss our observations and data reduction in Section 2, present our results in Section 3, discuss them in Section 4 and summarize in Section 5.

2. Observations and data reduction

2.1. Scanning FPI observations

The observations were held in the primary focus of the 6-meter telescope of Special Astronomical Observatory of the Russian Academy of Science (SAO RAS) with scanning Fabri-Perot interferometer mounted in universal focal reductor SCORPIO-2 Afanasiev & Moiseev (2011). Those observations and the parameters of the final images are listed in Table 1. In the table, T_{exp} is the exposure time in seconds, FOV is the field of view in arc minutes, $''/px$ is size of the pixel on the images in arc seconds, θ is the angular resolution in arc seconds, λ_c is central wavelength of the filters in Angstroms, and $\delta\lambda$ is spectral resolution (FWHM) in Angstroms.

2.2. Narrow-band imaging

We used narrow-band images of Hα, [O iii], [S ii] and corresponding continuum images obtained with NBI camera mounted on the 2.5-meter telescope of Caucasian Mount Observatory (CMO) of SAI MSU. We subtracted continuum and matched PSF across images to achieve uniform angular resolution. The paramerers of images are listed in the Table 1.

3. Analysis

3.1. Statistical moments and outflow measurements

To study the outflow, we plot the 0-th, 1-st and 2-nd statistical moments of our data cubes, which represent total flux, radial velocity and velocity dispersion of emission gas. We correct velocity fields for galaxy rotation (HI velfield) and systematic speed of Mrk 71, and correct line widths for instrumental and thermal broadening prior to calculating velocity dispersion. We also exclude regions with low signal-to-noise ratio from our analysis.

The outflow clearly separates from ambient ISM in velocity fields and does not outstand in line widths, except the very tip of the cone, where cone/ISM intensity ratio is close to unity, and observed line broadening can be accounted to the separation of the centroids of cone and ISM components in the velocity space. The position of the outflow in the velocity field coincides the elevated regions in line ratio maps without any observed decline to the outskirts, confirming the LyC leakage proposed by Micheva et al. (2017, 2019)

The velocity fields in Hα and [O iii], consistent with Ha measurements in van Eymeren et al. (2007, 2009), show rather slow, mildly accelerating outflow in the NW direction. Its velocities relative to ambient ISM are starting from 5 km/s at the origin of the cone and reach 30 km/s in the furthermost areas. However, the outflow does not outstand in surface brightness, which means the velocity field is strongly biased with the surrounding ISM, with surface brightness exceeding the outflow in the central areas.

3.2. Position angle and opening angle

We use the following procedure to extract the boundaries of the outflow: first, we extract contours in the velocity field (from v_{ISM} to $v_{out}/2$) with measure.find_contours from scikit-image (van der Walt et al. 2014), close them and choose the largest contour by area. Next, we split each contour into left and right side by the approximate position angle of the axis of the cone. Finally, for each contour we calculate the position angles of each point belonging to it relative to the cone apex, apply sigma-clipping, filter out contours without enough data points left and chose the contour with lowest dispersion in polar angle. We further refine the result iteratively, better accounting for beam smearing and right-left splits.

This approach in [O iii] is robust to parameter variation, yielding the position angles of the outflow edges $\theta_r = 30 \pm 1°, \theta_l = 90 \pm 1°$, providing the position angle of the axis $\theta_a = 90 \pm 1°$and the full projected opening angle $\theta_p = 60 \pm 2°$. The outflow does not outstand so well in Hα velocity field, and the same approach applied to Hα is rather unstable, providing roughly $\theta_p = 51 \pm 7°$.

3.3. Decomposition of the outflow

The line-spread function (LSF) of a FPI is a near-Lorentzian instrumental profile of fixed width. This allows us to fit Voigt profile with multiple components of fixed Lorentzian width and measure 3 parameters of each component: flux, radial velocity and line width (FWHM). We use sum of counts, weighted median and weighted MAD (median absolute distribution) as initial guesses respectively, and then iteratively fit the components. Bayesian Informatian Criterion (Schwarz 1978)is calculated between iterations to measure goodness of fit, and we use Moran's I statistics (Moran 1950) to analyse spectral distribution of the residuals.

We build a 2-component decomposition pipleline to decouple the outflow from ambient ISM. Hα and [O iii] spaxels

Table 1. Observational data

Data set	Date	T_{exp}, s	FOV	$''/px$	θ, $''$	Sp. range	λ_c, Å	$\delta\lambda$, Å
Fabri-Perot Interferometry								
FPI	2016 Oct 20	40 × 120	6.1′ × 6.1′	0.69	1.2	8.8 Å (Hα)	—	0.4
FPI	2017 Dec 14	30 × 120	6.1′ × 6.1′	0.69	1.4	5.1 Å ([O iii])	—	0.4
FPI	2022 Dec 31	40 × 150	6.1′ × 6.1′	0.69	1.8	8.8 Å (Hα)	—	0.4
Narrow-band imaging								
CMO NBI	2018 Feb 15,17	2100	11.1′ × 10.6′	0.155	0.9–1.2	[O iii] 5007 Å	4992	120.8
CMO NBI	2018 Feb 15,17	1500	11.1′ × 10.6′	0.155	0.9–1.2	[O iii] cont.	5117.8	94.6
CMO NBI	2018 Feb 15–21	1800	11.1′ × 10.6′	0.155	0.9–1.2	Hα	6559	76.5
CMO NBI	2018 Feb 15	600	11.1′ × 10.6′	0.155	0.8–1.0	Hα cont.	6426.9	122.6
CMO NBI	2018 Feb 21	2400	11.1′ × 10.6′	0.155	1.0-1.25	[S ii] 6717+31	6721.3	56.8
CMO NBI	2018 Feb 21	900	11.1′ × 10.6′	0.155	1.0-1.25	[S ii] cont.	6959.7	153

in the outflow show thin profiles (no broadening compared to the surrounding ISM), moderately asymmetric in blue wings. The velocity difference between the outflow and ambient ISM is close to the spectral resolution of our observations, which brings 2-component decomposition into the degeneracy between 'amplitude' and 'velocity', making profiles effectively unresolvable. To address this issue, we fix the velocity of the ISM component on the value measured for the ambient ISM, and fit a joint profile with the decomposition above. With S/N ratio >10 two-component decomposition is stable even without refitting, but the residual sky subtraction artifacts, which have relative amplitudes about 5% in stacked profiles, prevent analysis of weaker components.

We use two toggles in our algorithm: 'norm' and 'shift'. First option normalizes all profiles before stacking, making them weighted by area instead of intensity, raising significance of weaker areas. Second option corrects centroid shift of each profile before stacking, potentially reducing beam smearing: amplitude-velocity degeneracy with fixed line width is particularly sensitive to it. However, it introduces systematic errors, especially in low-SN areas, which grow when both options are used simultaneously.

(не знаю, насколько подробно это надо писать, тут у меня построены радиальные и азимутальные профили лучевой скорости из декомпозиции по 3 бинам каждый. Они показывают, что аутфлоу в среднем немного ускоряется, но в целом имеет более-менее стабильную лучевую скорость 25-30 км/с. Также он заметно замедляется к краю, что можно объяснить нагребанием МЗС. И вообще тут куча работы проделана, потому что я долго думал, что главное – нормально померить лучевую скорость). Но вроде особой научной ценности эти измерения не представляют.

3.4. Reconstructing the 3D geometry of the outflow

3.4.1. Inclination to the image plane

Direct measurements of inclination of such outflows is usually impossible, but we have several means to estimate and constrain its value. Note that we find the cone lying relatively close to the image plane, so we refer to the inclination angle as the angle between the outflow axis and the image plane (not to be confused with LOS angle).

First, outflows tend to align with minor axis of the galactic disc, and in Mrk71 we observe angle 15 degrees between the cone axis and the minor axis of the disc in the image plane. Inclination of the disc is well measured and is 27 degrees to the image plane, which implies our simplest prior $i_{inc} \simeq 27° \pm 15°$ degrees.

Next, we do expect the cone to expand freely after breakout with $v_{perp} \approx c_s$ which brings us to Mach cone geometry assumed by In this toy model we can tie measured radial velocity, projected opening angle and sound velocity which can be obtained via T_e measurements (Yarovova et al. 2025) to infer inclination angle. Adopting it, we get $i_{Mach} \simeq 36.5° \pm 9°$, which is relatively close to the prior estimate. This means Mach geometry is close to reality, while there is no simple physical model or law that could predict Mach opening angle for an outflow. In jet propulsion theory, supersonic jets can have a broad variety of opening angles, depending on the pressure ratio.

Finally, looking at HI-Ha maps from we can see that after a confined cone-like phase the outflow opens broadly into CGM at high disc heights. Estimating the disc scale height at $h_0 \simeq 500pc$ () and the visible span of the pressure confined cone as $400pc$, we expect an opening to occur around 1.5-2 scale heights (momentum-driven shell), which keeps us in the moderate inclination range ($> 20°$) from the minor axis.

Here we adopt $i_{pl} \simeq 34 \pm 8$ degrees from the image plane as a joint measure from first and second approaches.

4. Discussion

The observed parameters of the outflow appear to be moderately inconsistent with the luminosity of Mrk71B cluster powering it.

4.1. Age discrepancy of SSC Mrk71B and its outskirts

Prior measurements have demonstrated a very young, dense core ("Knot B", 2.5 Myr, 40 O + 800 B stars) and a moderately older (3-5 Myr) surrounding association ("Cluster B" with similar star count (~60 B2+ stars around)) Это Насте бы дать пересчитать с современными моделями массивных звёзд, но можно в целом и так оставить. No RSG are present in the surroundings, implying absence of significant population >7 Myr. No signs of SN activity have been observed, except from a variable point-like

X-ray source in knot B () which is likely a compact accreting object. The strongest of those signs is flat thermal radio spectrum in knot B () which means SN rate < 10^{-5}yr^{-1}. This pushes inferred stellar ages of B association strongly below 4 Myr, and significant presence of WR population around (3 confirmed + 3 candidates - Настя проверь?) lands most probable age estimate into 3.5 ± 0.5 Myr window. We will adopt this estimate as our best guess.

4.2. Energy and momentum budget of the knot B

$\sim 10^{51}$ erg lifetime, long calculations here (взять из starburst99) $\sim 10^{44}$ g sm/s lifetime, long calculations too We model stellar population as 2.5 Myr core with 3.5 Myr outskirts, comparable masses. Accretion to a compact object could provide a lot of power, but is limited with its companion mass and mass losses during evolution. We find possible accretion effect minor compared to stellar luminosity, while excluding exotic scenarios such as multiple systems.

4.3. Budget comparison

Supply « Demand Плотность из SII ratio это плотность самых плотных клампов

надо взять плотность HI

4.4. The blowout morphology

The observed spatial extent (>900 pc) clearly indicates that the blowout could have formed in reasonable times (1-2 Myr) only with fast (>1000 km/s) wind. The slow optical phase likely is

Furthermore, Roy et al. (1991) measured radii of the cavern surrounding knot B. They reported diameters 18 and 28 pc in [O ii] and ??? lines. In the standard picture by , a pressurized bubble keeps growing until it reaches significant pressure gradient in the gaseous disc, and starts accelerating in the direction of the gradient. This happens usually when a bubble reaches at least 1 scale height and causes bipolar outflows. In case of Mrk 71B, which has unusually thick disc with strong flaring (scale height varies from 0.4 kpc in the inner regions to 0.8 kpc in the outskirts, $\sim 0.5\pm0.1$ kpc at Mrk71 radius), bubble-to-height ratio becomes <0.1, which strongly advices against pressure-driven blowout. Possible explanations include either early leakage through luckily positioned low-density channels or a strongly anisotropic collimated feedback from accretion on the compact object.

4.5. Venting through chimneys

In this setup a hot fast superwind (M>10) finds a narrow low-pressure channel early and begins venting into the CGM, suppressing bubble growth. The cone walls are slowly pushed in the perpendicular direction, bringing opening angle closer to the currently observed 60 degrees, and the entrained material is slowly accelerated

5. Summary

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